

J.A.P. Small Wind Turbine

Objective: To invent a resilient and affordable energy for all.

Bangkok, 25 November 2023

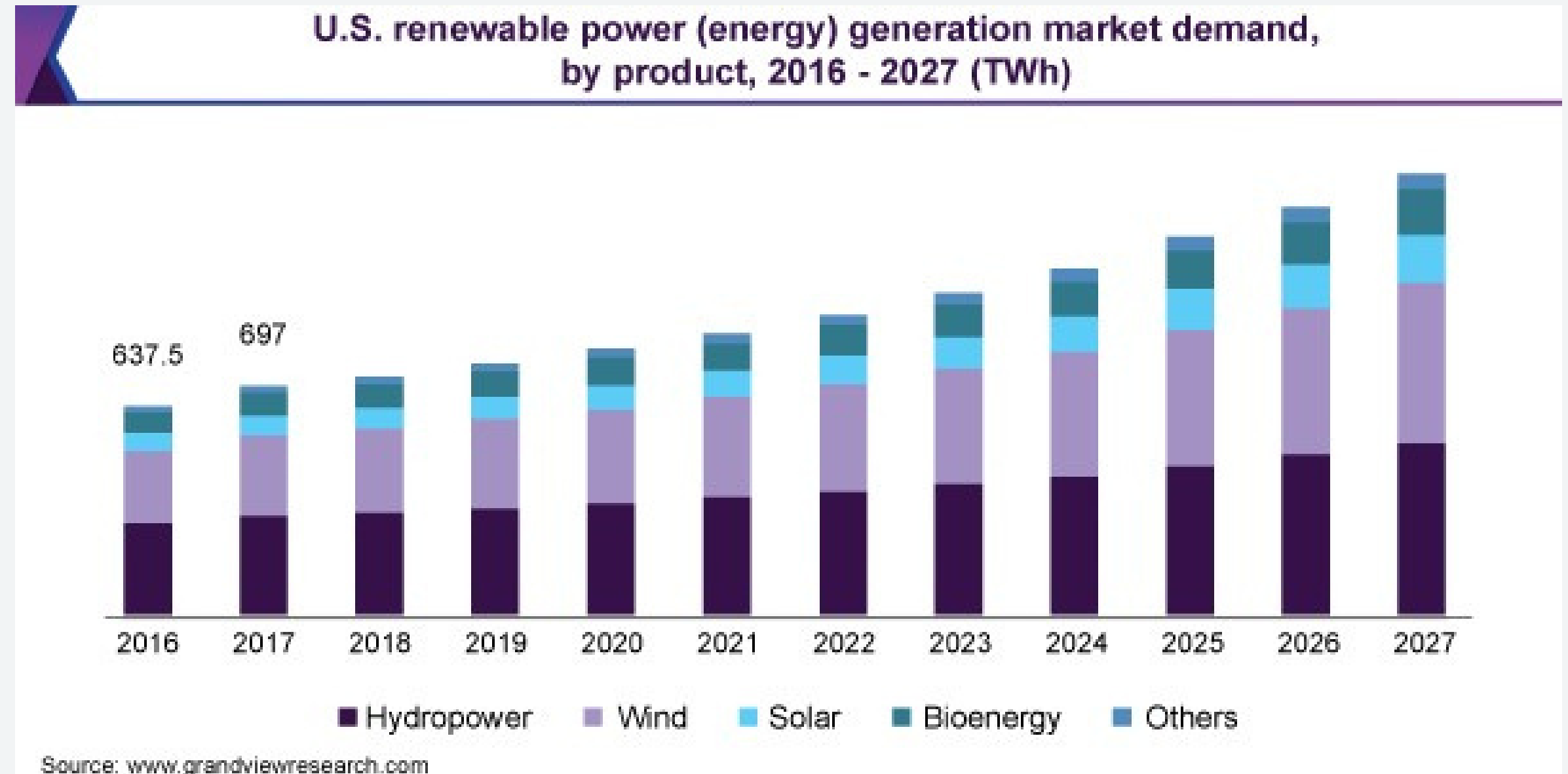
TEAM J.A.P.

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WHAT IS HAPPENING?

**Renewable
energy
is in need.**



SITUATION ANALYSIS

JAP TURBINES

IMPACT

WHAT IS HAPPENING?

Renewable energy is beneficial for our future.

- In 2025, up to 20% ITC bonus will be allocated for wind turbine projects located in low income communities.
- New IRA provisions by REAP doubled the maximum allowance grant size to \$1 million on cost for renewable energy projects.



SITUATION ANALYSIS

JAP TURBINES

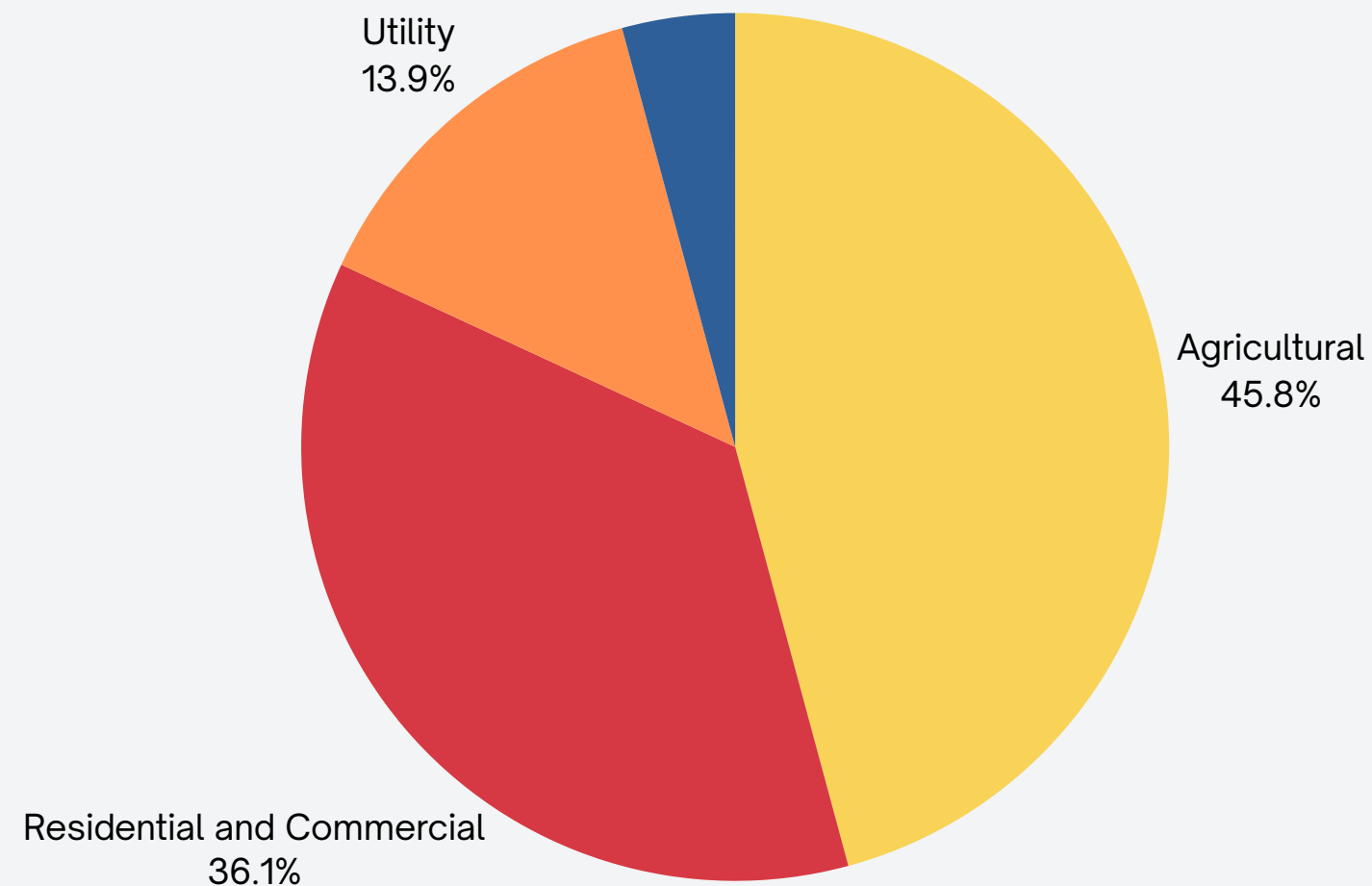
IMPACT

WHAT IS HAPPENING?

Customer Segment

>>>>>>

Ultimate Goal



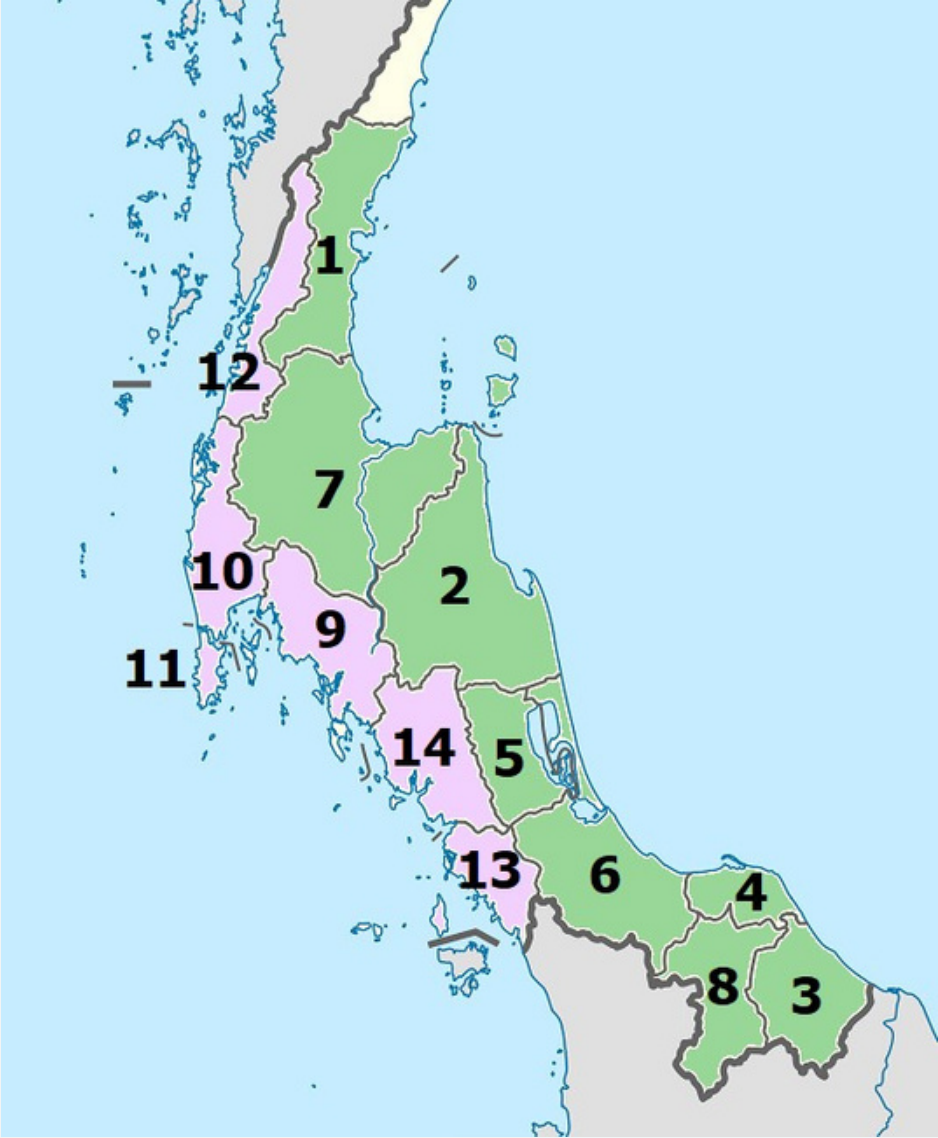
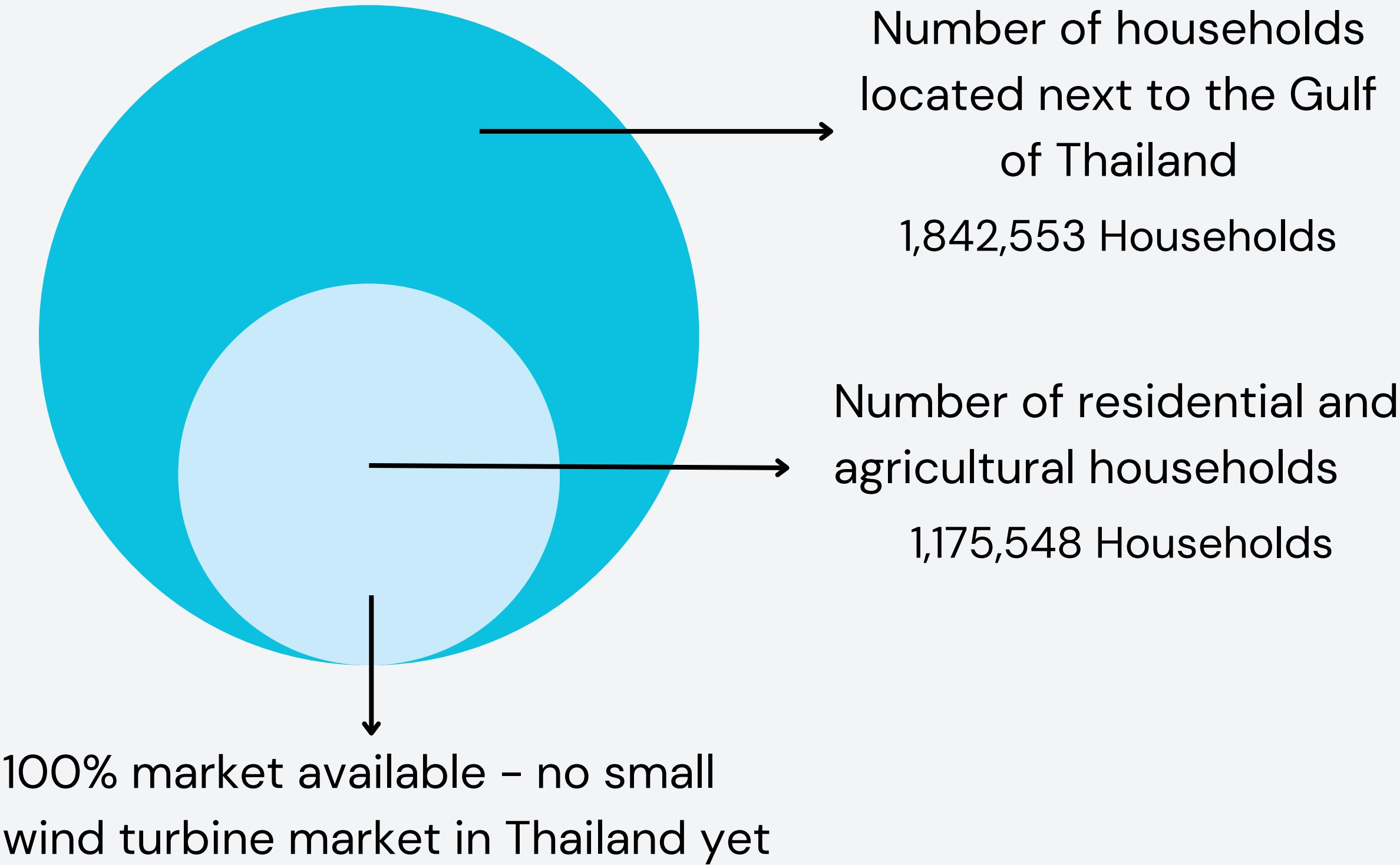
To gain more revenue,
JAP Turbine focuses on
agriculture and
residential customers

SITUATION ANALYSIS

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Going South



Areas surrounding the Gulf of Thailand (green) have biggest potentials of wind (6-7 mph)

J.A.P. TURBINE PROFILE

AERODYNAMICALLY DESIGNED BLADES

THREE BLADES ROTOR - 3 METERS HEIGHT

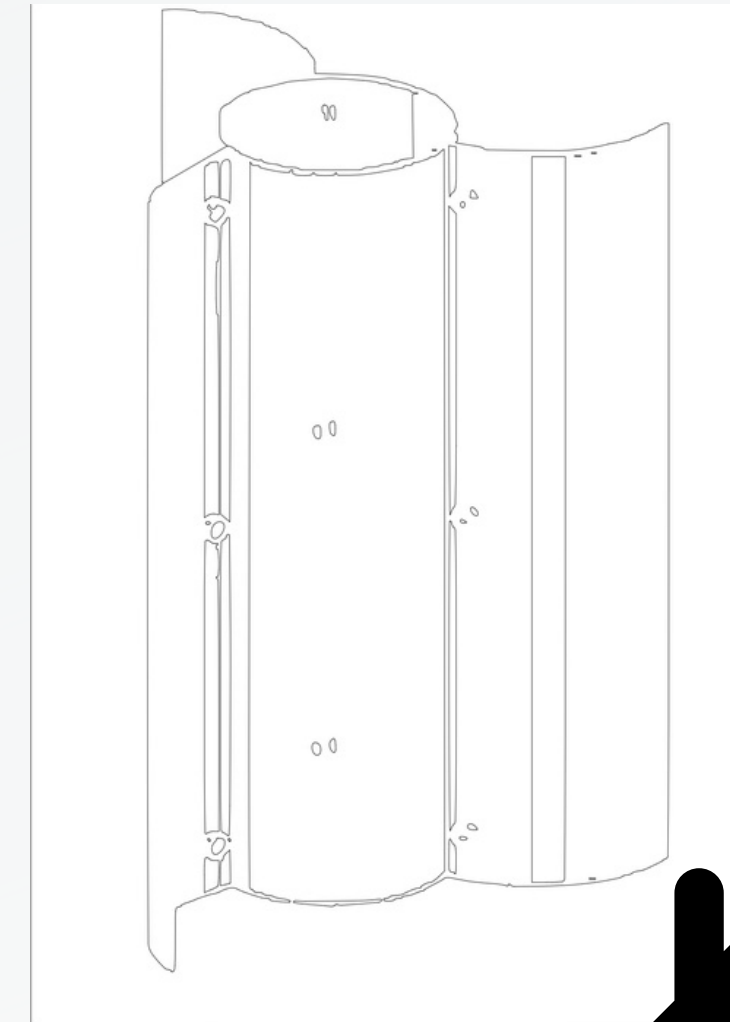
ENVIRONMENTAL FRIENDLY MATERIALS



AT LEAST 2.5M/S WINDSPEED REQUIRED

ACCEPT WIND FROM ANY ANGLES

TRANSFORMER TO INCREASE EFFICIENCY

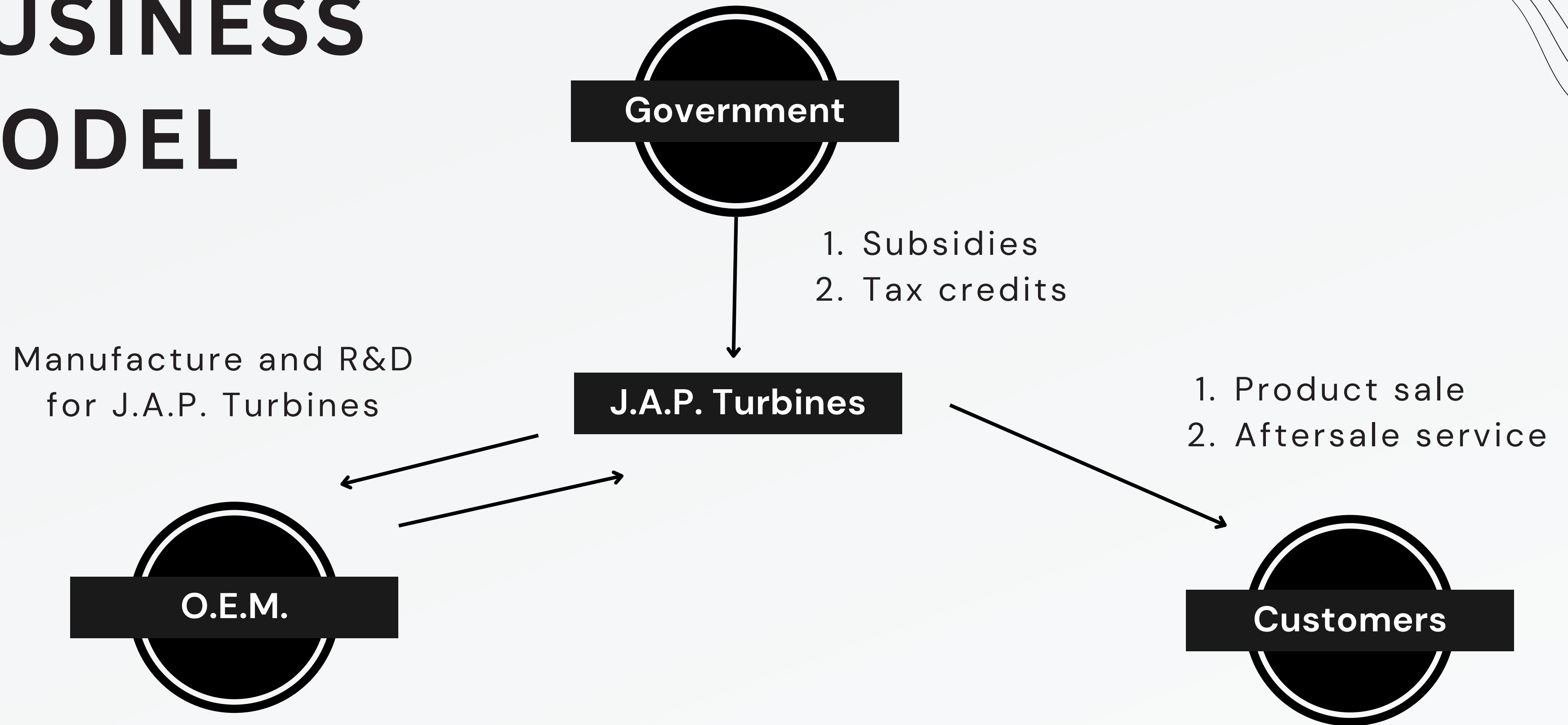


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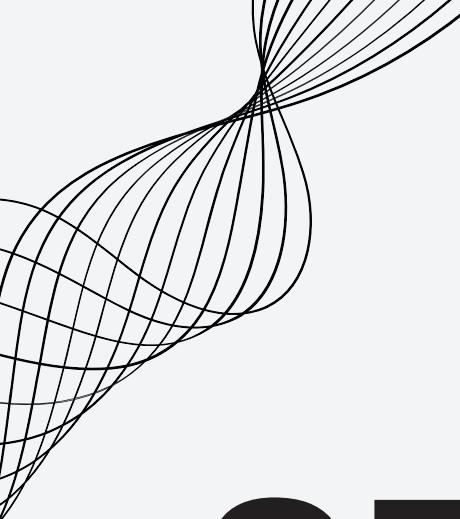
BUSINESS MODEL



SITUATION ANALYSIS

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MARKET STRATEGIES

... to build **strong
relationship** with individual
households through
customer service and
after-sale service.

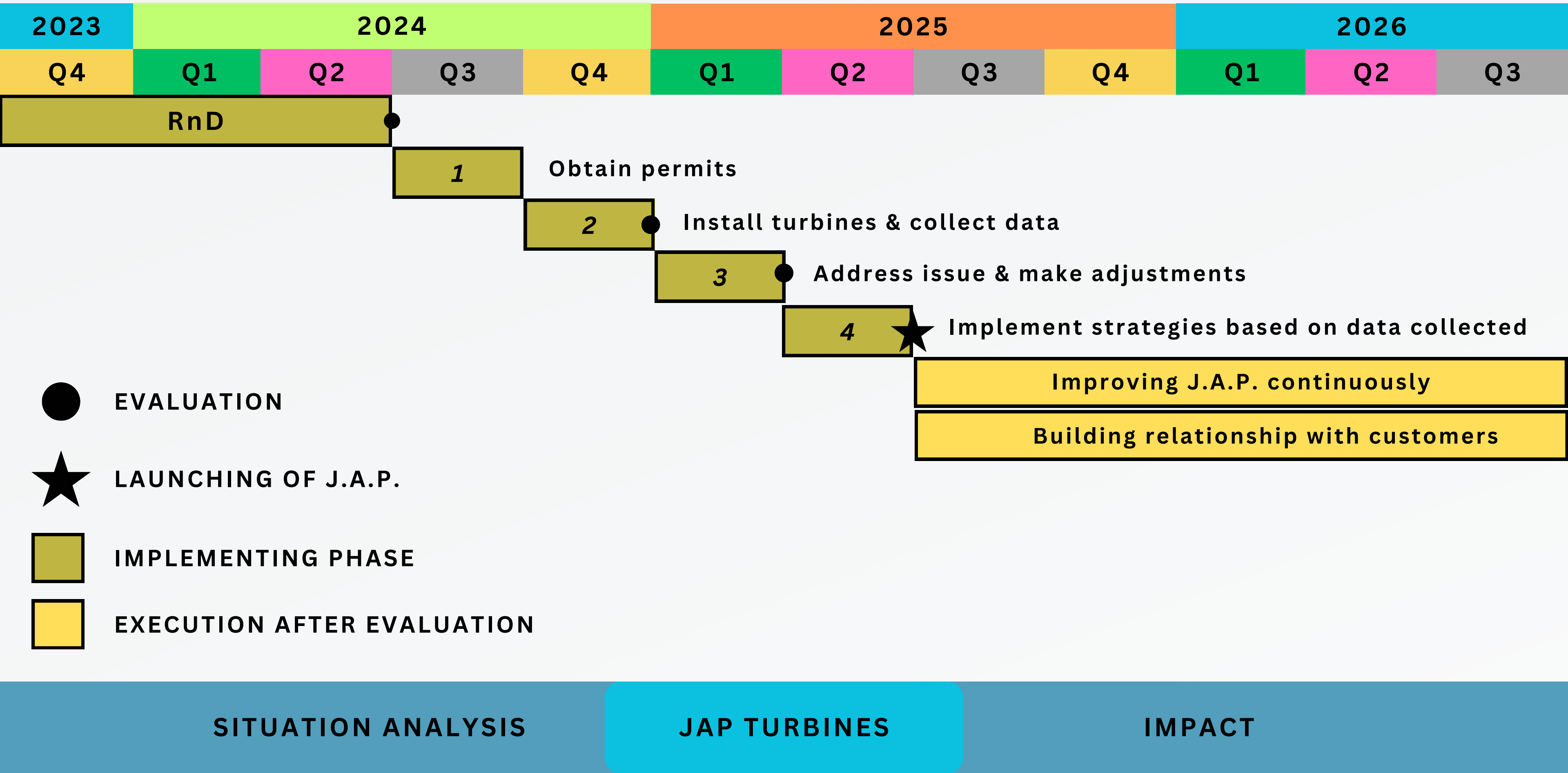


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IMPLEMENTATION



OUR TURBINE AND SDG



Ensure access to
**affordable, reliable,
sustainable and
modern energy for all**



Build resilient
**infrastructure, promote
inclusive and sustainable
industrialization and
foster innovation**



Take urgent action
**to combat
climate change
and its impacts**

Offering customers
an eco-friendly
energy source at
lower price.

Made out of
environmental
friendly materials.

Bringing a vision to
push wind as
renewable and more
efficient energy.

SITUATION ANALYSIS

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IMPACTS

on Global

**More solutions for
climate change and
solve the electricity
shortage problems**

on Investor

**Satisfy on investing in
a sustainable and eco-
friendly future**

on Society

**Wind turbines can be
affordable to all people**

on Customer

**Customers saving their
cost for their
electricity bills and
more eco-friendly
lifestyle.**

DISCLAIMER

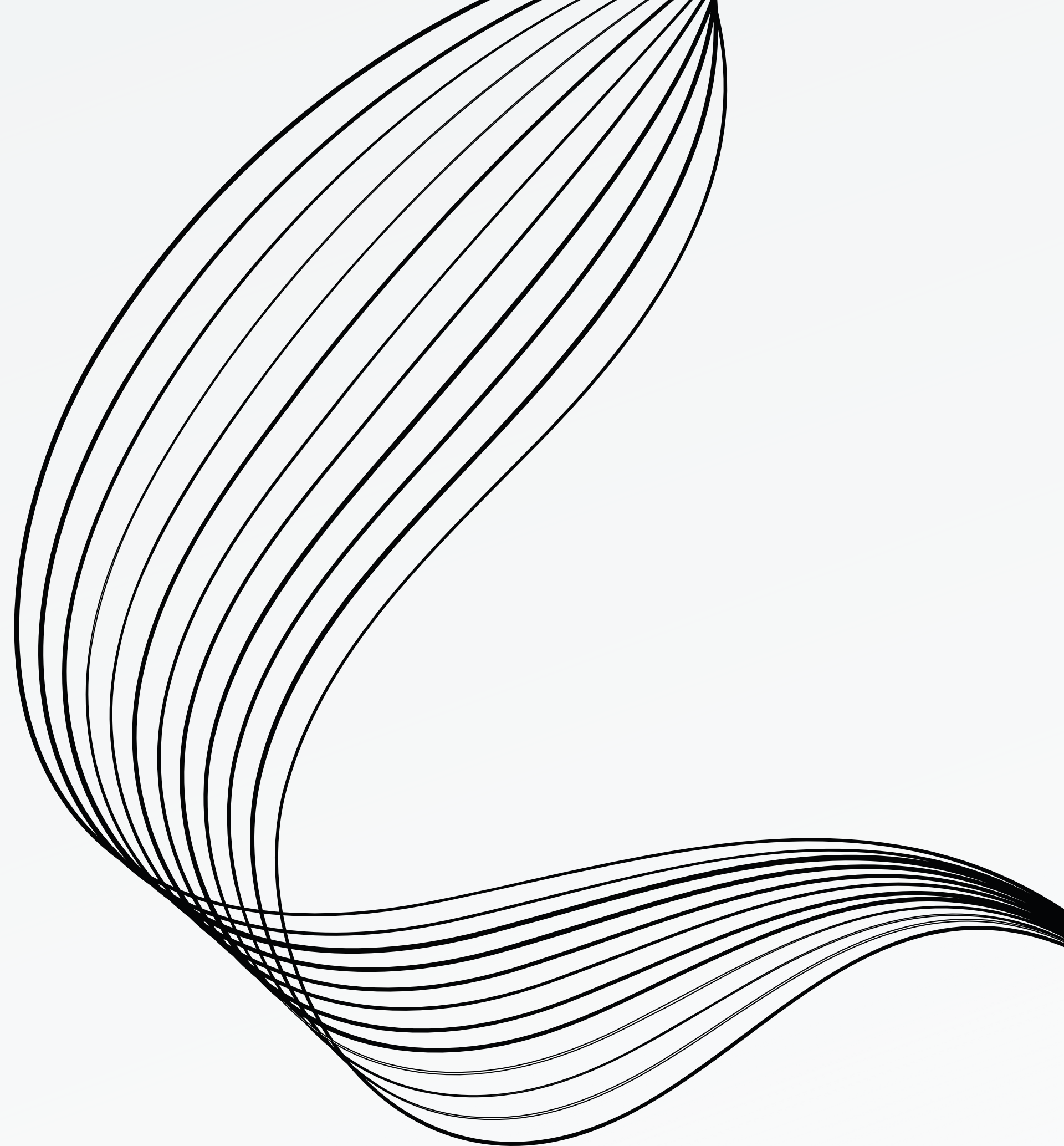
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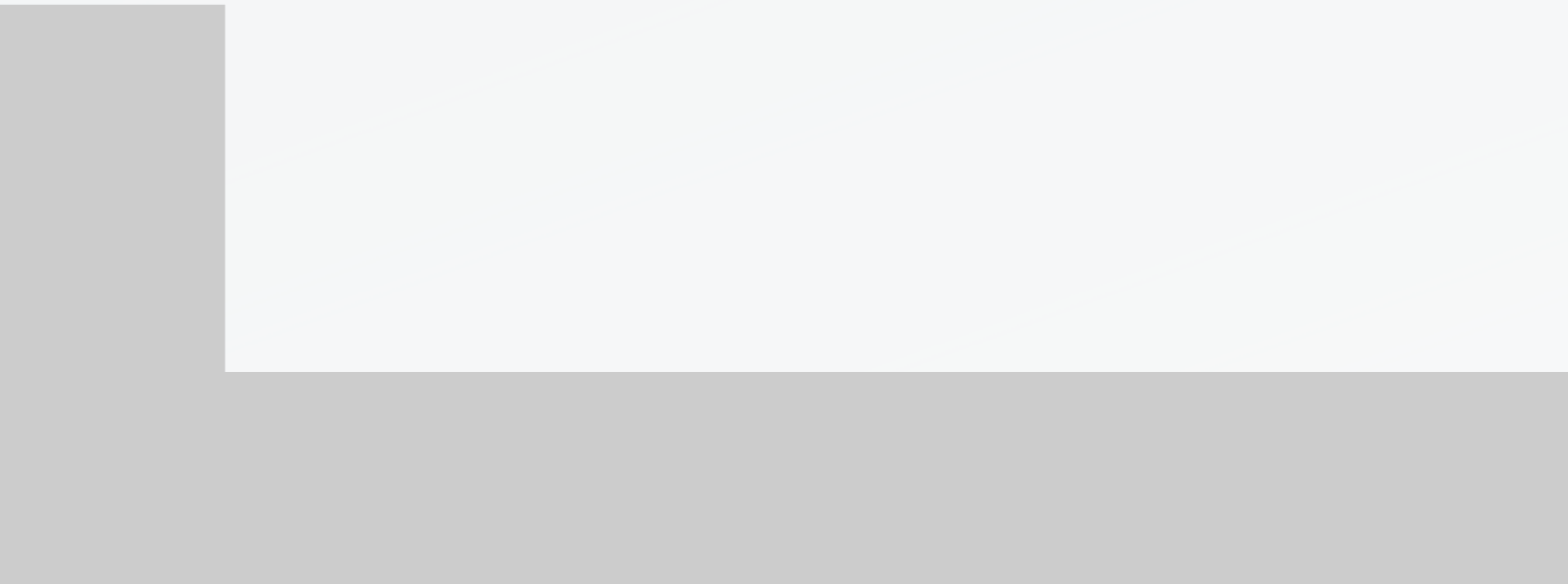
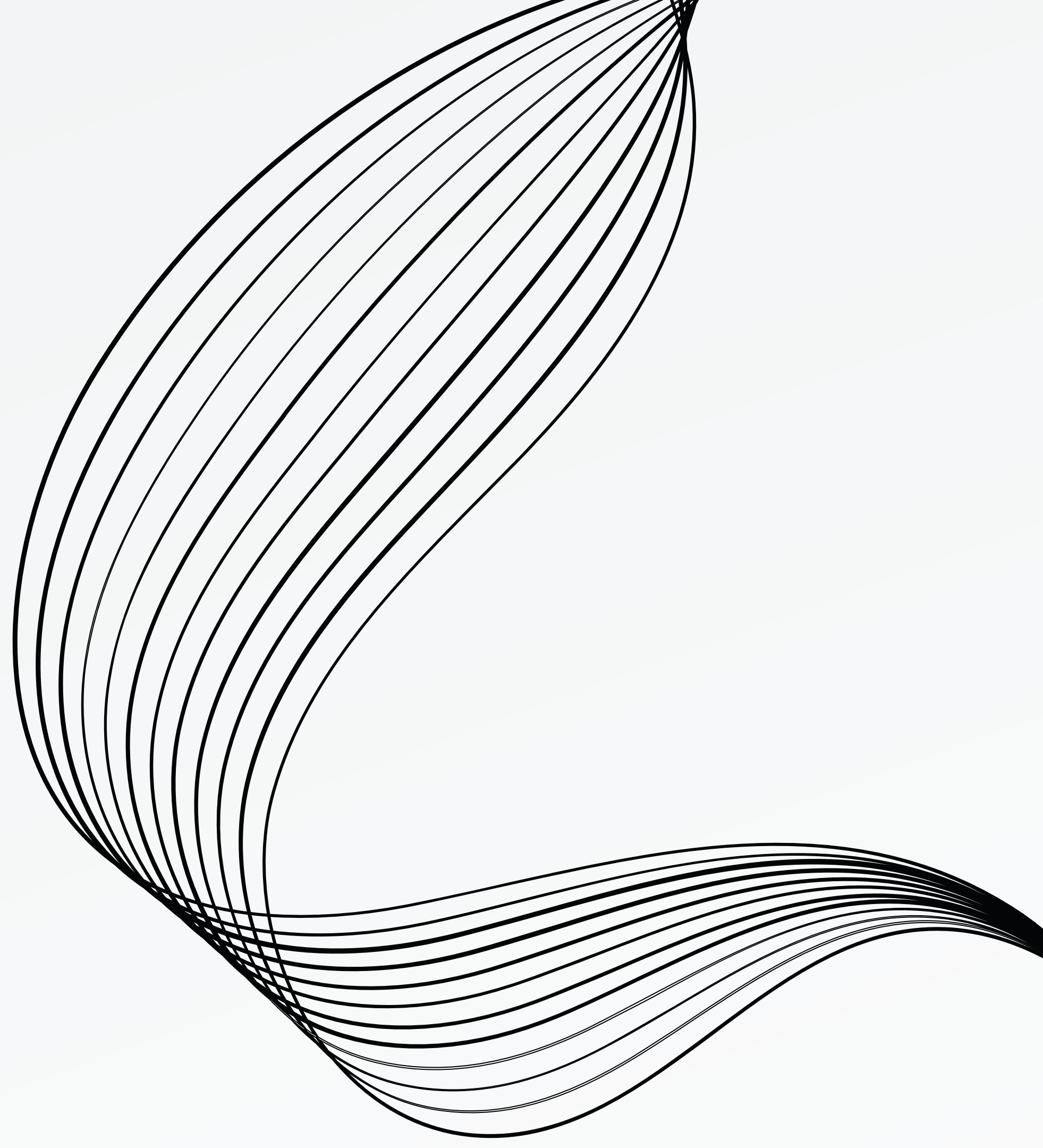
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THANKS FOR WATCHING

We're ready for Q&As.

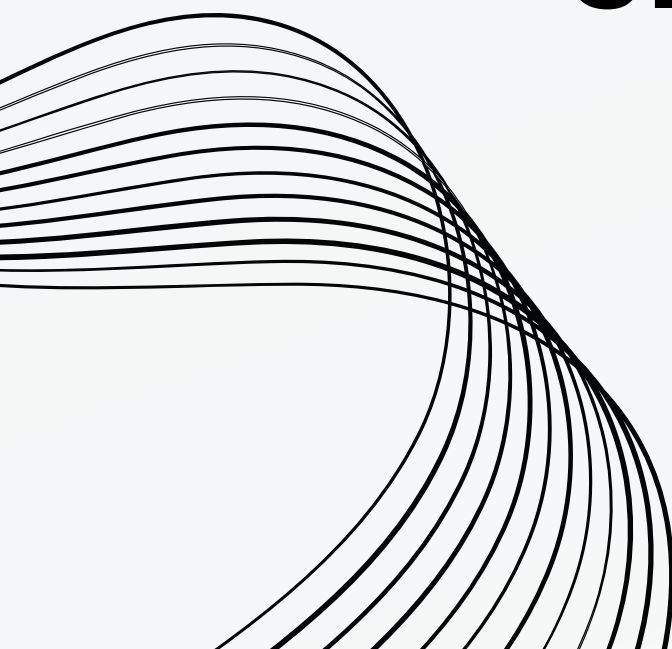


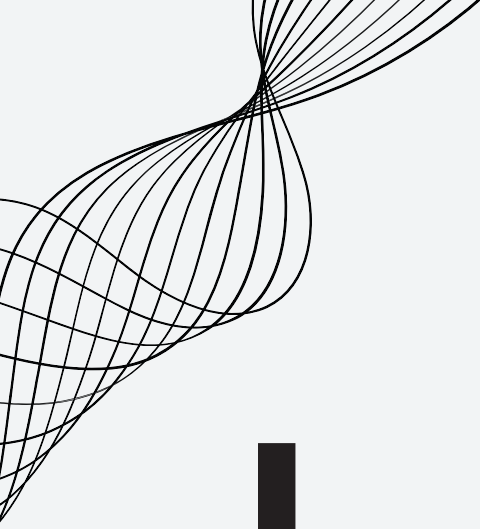
APPENDIX



**Comparison
between
wind energy and
other renewable
energies.**

Wind Energy	Other Renewable Energy (solar, hydro, etc.)
Widely available resource	Resource is not widely available
Quick installation	Complex and time consuming construction and installation
Able to coexist with city life, agriculture, and ecosystem	May impact the ecosystems





J.A.P. TURBINES

What differs us
from others?

J.A.P. Turbines	Other Wind Turbines
Virtually silent during operation	High noise level
Easy to install and maintain	Complex to install and maintain
Operate even in low wind speed	Only operate once the wind stream reached certain speed
Accept wind from any angles	The rotor must face the wind stream

SITUATION ANALYSIS

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RISK MANAGEMENT

Natural Disasters

- Risk? Damage from extreme weather events, eg: hurricanes and earthquakes
- How to beat?
 1. Designing turbines to withstand local weather conditions
 2. Considering insurance coverage for natural disasters

Policy & Regulatory

- Risk? Changes in government policies affecting wind energy
- How to beat?
 1. Staying informed about regulatory changes
 2. Adjusting project plans accordingly
 3. Advocating for stable & supportive policies

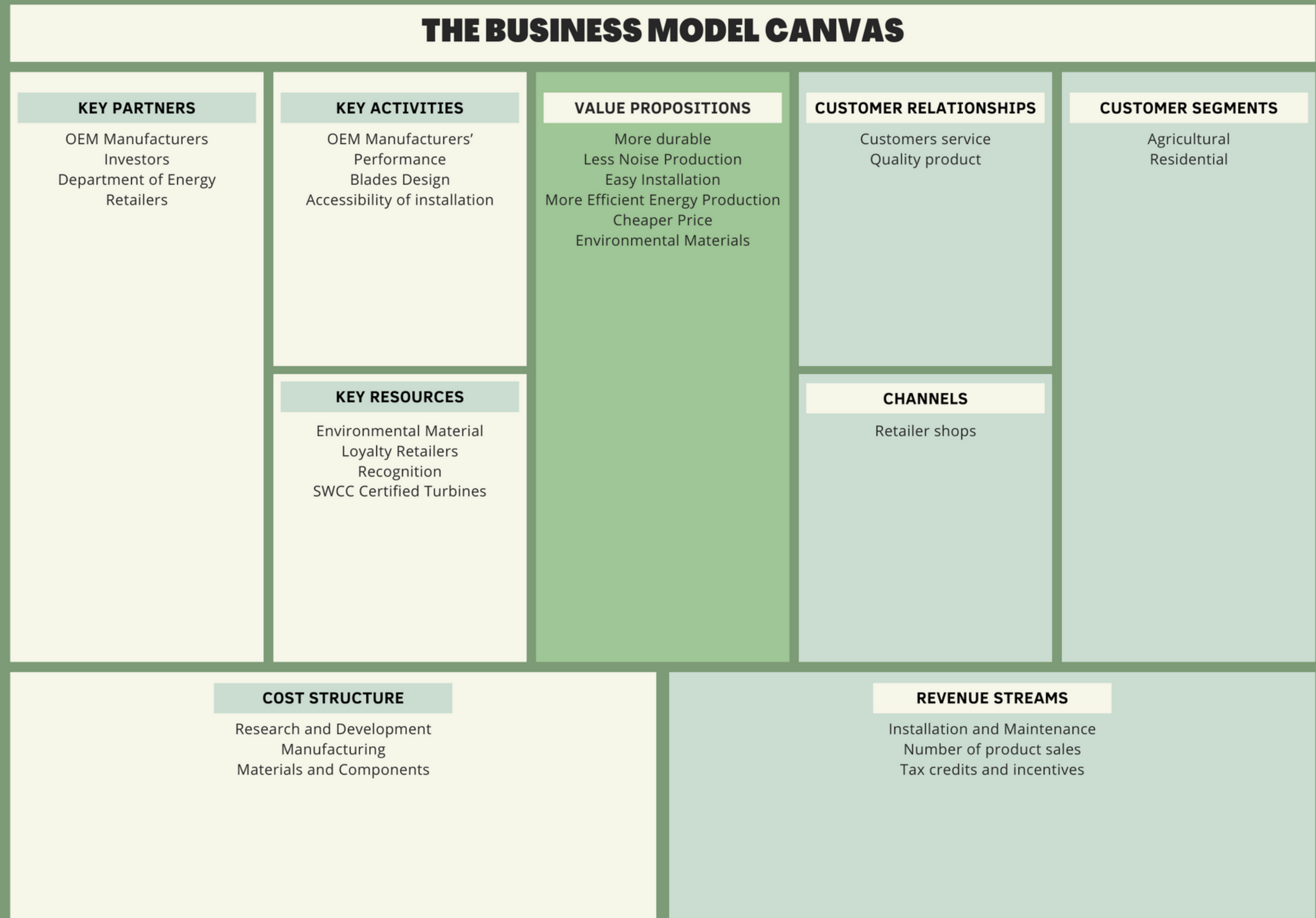
Supply Chain Issues

- Risk? Disruption in the supply chain for turbine components
- How to beat?
 1. Diversifying suppliers when possible
 2. Maintaining good relationships with suppliers
 3. Preparing contingency plans for unexpected disruptions

Wind Resources Variability

- Risk? Inconsistent or lower-than-expected wind speeds impacting energy production
- How to beat?
 1. Conducting a thorough wind resource assessment before the project
 2. Implementing energy storage system to help buffering fluctuations in energy production

BUSINESS MODEL CANVAS



NUMBER OF PROJECTS ABSORBED BY CUSTOMER'S SEGMENTS

	2014	2015	2016	2017	2018	2019	2020	2021	2022
	Percentage of Number of Projects (%)	Percentage of Number of Projects (%)	Percentage of Number of Projects (%)	Percentage of Number of Projects (%)	Percentage of Number of Projects (%)	Percentage of Number of Projects (%)	Percentage of Number of Projects (%)	Percentage of Number of Projects (%)	Percentage of Number of Projects (%)
Customer Type									
Agricultural	30.5	45.1	27.0	22.7	10.7	30.1	28.9	59.3	33.3
Residential	35.9	23.8	33.0	37.3	17.9	30.1	13.3	15.3	25.6
Commercial	9.9	6.6	15.0	10.7	7.1	16.4	8.9	10.2	25.6
Industrial	3.8	4.9	2.0	4.0	10.7	6.8	6.7	5.1	2.6
Institutional	9.2	9.0	10.0	4.0	10.7	5.5	4.4	1.7	2.6
Government	6.9	9.0	7.0	9.3	14.3	5.5	31.1	3.4	0.0
Utility	3.8	1.6	6.0	12.0	28.6	5.5	6.7	5.1	10.3
	2014	2015	2016	2017	2018	2019	2020	2021	2022
Total Number of Projects	131	122	100	75	28	73	45	59	39

Source: Distributed Wind Market Report: 2023 Edition, Office of Energy Efficiency & Renewable Energy, US Department of Energy